

Deep Learning for Organoid Analysis: Graph-Based Modeling of 3D Cellular Architectures

Alexandre Martin

INRIA Sophia-Antipolis – Morpheme Team
In collaboration with IPMC Nice & Paris Cité University

ANR Morpheus Project

Thesis Defense



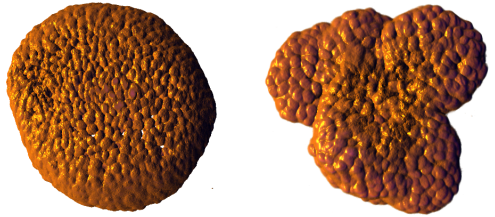
Organoids: Mini-Organs in a Dish

What are organoids?

- 3D structures grown *in vitro* from stem cells
- Self-organize to mimic real organ architecture
- Bridge gap between 2D cultures and animal models

Applications:

- Personalized medicine
- Drug screening
- Disease modeling



The Challenge: Automated Phenotype Classification

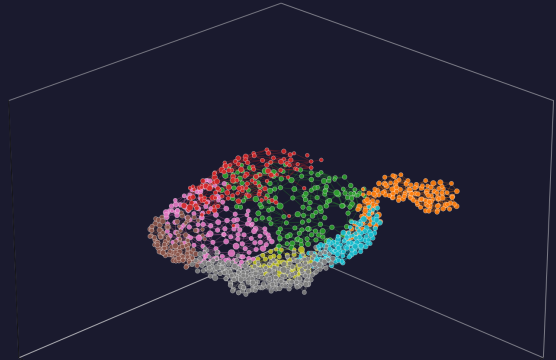
✓ Cystic (Healthy)

- Spherical, central cavity
- Organized structure
- **Uniform** cell distribution

⚠ Cauliflower (Perturbed)

- Irregular, multiple buds
- Disorganized structure
- **Clustered** cell distribution

Cauliflower Organoid - Cluster Analysis
(Hierarchical Clustering)



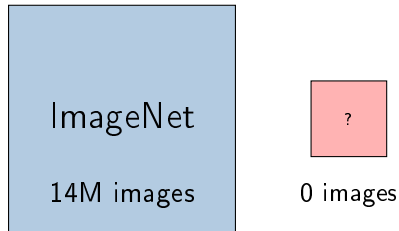
Starting Point: No Annotated Data

At the beginning of this thesis:

- ✗ No labeled organoid dataset
- ✗ No public benchmarks
- ✗ Expert annotation = expensive

Key question:

How to develop and validate methods without real data?



Our Approach

Synthetic data + Graph modeling
to bootstrap the research

Three Key Challenges

Challenge 1

Manual Analysis

- 30 min/organoid
- Subjective
- Not reproducible

Challenge 2

Data Scarcity

- No public datasets
- Expensive annotation
- Domain-specific

Challenge 3

3D Geometry

- No preferred orientation
- 2 GB per organoid
- Rotation invariance

Need

Automated, data-efficient, geometrically robust methods

Q1 Can we model organoids as geometric graphs?

- Capture biological information while compressing data

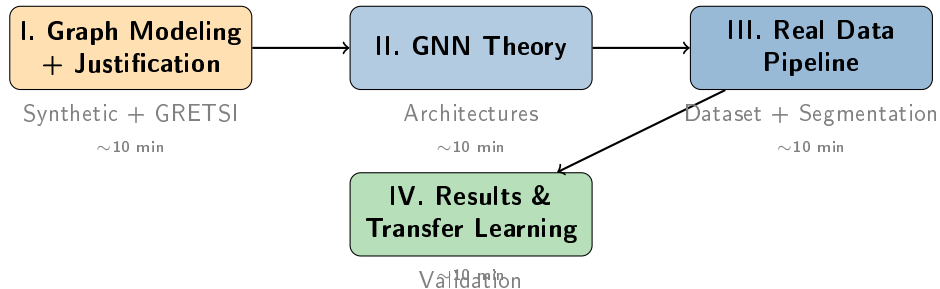
Q2 Do GNNs outperform classical spatial statistics?

- Justify the deep learning approach

Q3 Can synthetic data enable transfer learning?

- Compensate for lack of real annotations

Presentation Outline



Narrative: No data → Synthetic + theory → Real data arrives → Transfer learning

Part I

Graph Modeling & Justification

Why graphs? Why GNNs?

Why Model Organoids as Graphs?

Key Biological Insight:

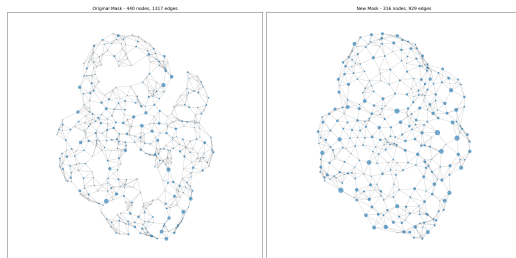
Cell **spatial organization** differs between phenotypes

Graph Abstraction:

- Each **cell** → **node**
- Spatial proximity → **edges**
- Node features: position (x, y, z) , volume

Advantages:

- **1000× compression** (GB → MB)
- Captures relational structure
- Natural for point cloud data



Synthetic Organoid Generation: Motivation

Problem: No real data at thesis start

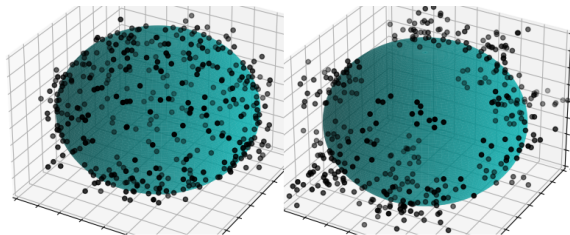
Key Biological Insight:

Cell distribution differs between phenotypes

- Cystic → **uniform** distribution
- Cauliflower → **clustered** distribution

Solution: Spatial point processes

- Mathematical models for point patterns
- Controllable parameters
- Perfect ground truth labels



Uniform vs clustered distributions

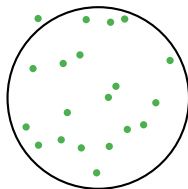
Point Processes: Mathematical Foundation

Poisson Process (Cystic)

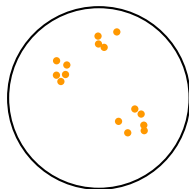
Intensity: λ (points per unit volume)

$$P(N(B) = k) = \frac{(\lambda|B|)^k e^{-\lambda|B|}}{k!}$$

Complete spatial randomness



Poisson (uniform)



Matérn (clustered)

Matérn Cluster (Cauliflower)

Parameters:

- κ : parent intensity
- μ : children per parent
- r : cluster radius

▶ `point_processes.gif`

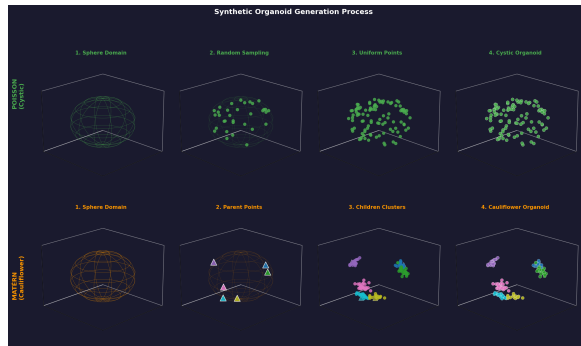
Synthetic Dataset: Statistics

Generation Parameters:

- Cell count: 100–500 per organoid
- Sphere radius: normalized to 1
- Volume features: log-normal distribution

Dataset Splits:

Split	Count
Training	70,000
Validation	15,000
Test	15,000
Total	100,000



Generation pipeline steps

Advantages:

- Perfect ground truth
- Unlimited supply
- Controllable difficulty

Research Question:

When should we prefer GNNs over classical methods?

Classical Approach:

- Ripley's K, F, G functions
- Random Forest classifier
- Well-established in spatial statistics

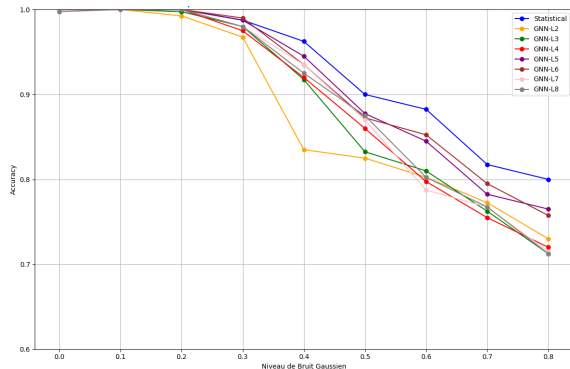
Deep Learning Approach:

- Graph Neural Networks
- Learn features from data
- End-to-end learning

Protocol:

- Synthetic data with controlled ground truth
- Test noise robustness and geometric generalization

Result 1: Noise Robustness



Observation:

Spatial statistics are **more robust to noise**

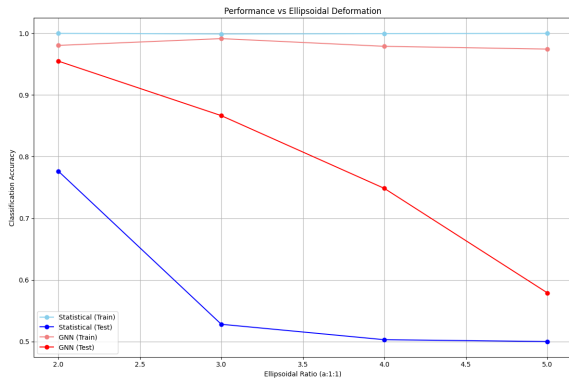
Why?

- Ripley's functions average over many point pairs
- Natural smoothing effect
- GNNs can propagate noise through layers

Optimal GNN depth: 5-6 layers

- Deeper → over-smoothing

Result 2: Geometric Generalization



Experiment:

Train on spheres, test on ellipsoids

Results:

- Spatial stats: 100% $\rightarrow$ 65% (-35%)
- GNNs: 95% $\rightarrow$ 82% (-13%)

Why?

- Ripley's K assumes isotropy
- GNNs learn **topological** patterns

Key Finding

GNNs excel when morphologies vary — **exactly our situation with real organoids!**

Justification: Why GNNs for Organoids

GNNs Excel When:

- Morphologies vary
- Shapes are irregular
- Geometry is non-spherical

⇒ **Real organoids**

Spatial Statistics Excel When:

- Geometry is perfectly spherical
- Consistent across samples
- Low noise environment

⇒ **Idealized conditions only**

Foundation for Our Approach

For cystic vs cauliflower with variable morphologies, **GNNs are the appropriate choice**

Part I Summary: Justification

- ① **Graphs** naturally capture cellular spatial organization
- ② **Synthetic data** enables method development without annotations
- ③ **GRETSI study** demonstrates:
 - GNNs > spatial statistics for variable morphologies
 - Justifies our deep learning approach

Next

Now that we've justified GNNs, let's understand how they work

Part II

Graph Neural Networks

Theoretical Foundations

Why Not 3D CNNs?

Memory Constraints:

- Single organoid: ~ 2 GB
- $2048 \times 2048 \times 200$ voxels
- GPU memory: prohibitive

Downsampling Issues:

- Destroys cellular details
- Individual cells unresolvable

No Native Invariance:

- Not invariant to 3D rotations
- Requires extensive augmentation

Grid-Based Limitation:

- Treats organoid as voxel grid
- Doesn't model cells as entities

Conclusion

Graphs + GNNs overcome all these limitations

GNNs: Message Passing Paradigm

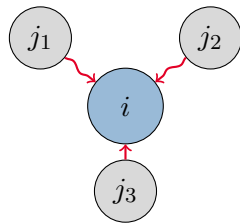
Core Idea: Nodes exchange information

Update Rule:

$$h_i^{(l+1)} = \text{UPDATE} \left(h_i^{(l)}, \text{AGG} \left(\{h_j^{(l)}\}_{j \in \mathcal{N}(i)} \right) \right)$$

Iterative process:

- 1 Each node sends messages to neighbors
- 2 Aggregate incoming messages
- 3 Update node representation



Messages flow to central node

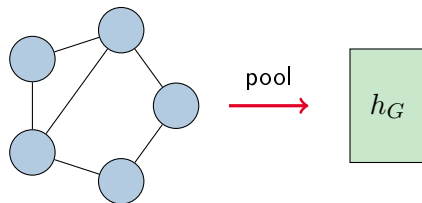
🔴 message_passing.gif

From Node to Graph: Global Pooling

Goal: Graph-level prediction (classification), not node-level

Pooling Methods:

- **Mean:** $h_G = \frac{1}{|V|} \sum_{i \in V} h_i$
- **Max:** $h_G = \max_{i \in V} h_i$
- **Sum:** $h_G = \sum_{i \in V} h_i$
- **Attention:** learned weights



Our Choice: Mean + Max concatenation

Classification:

$$\text{Nodes} \xrightarrow{\text{GNN}} \text{Embeddings} \xrightarrow{\text{Pool}} h_G \xrightarrow{\text{MLP}} \hat{y}$$

GCN: Graph Convolutional Network (Baseline)

Kipf & Welling, 2017

Update Rule:

$$h_i^{(l+1)} = \sigma \left(\sum_{j \in \mathcal{N}(i) \cup \{i\}} \frac{1}{\sqrt{d_i d_j}} W^{(l)} h_j^{(l)} \right)$$

Properties:

- Normalized mean aggregation
- Simple and effective baseline
- ✗ Treats all neighbors equally
- ✗ No distance/importance weighting

GAT: Graph Attention Network

Veličković et al., 2018

Key Idea: Learn attention coefficients to weight neighbors

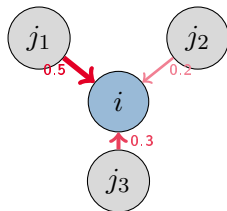
1. Compute attention scores:

$$\alpha_{ij} = \text{softmax}_j (\text{score}(h_i, h_j))$$

2. Weighted aggregation:

$$h_i^{\text{new}} = \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} \cdot W h_j \right)$$

✓ **Best performance** in our experiments



Learned attention weights

Zaheer et al., 2017

Approach: Treat organoid as unordered set of cells (ignore edges)

$$f(X) = \rho \left(\sum_{x \in X} \phi(x) \right)$$

Properties:

- Permutation invariant
- ✗ Ignores local spatial structure
- Each cell processed independently before aggregation

Observation: Performs reasonably well

⇒ Global statistics are informative, but **local structure helps**

Geometric Equivariance: Why It Matters

Problem:

Organoids have no preferred orientation in 3D culture

Invariance (for classification)

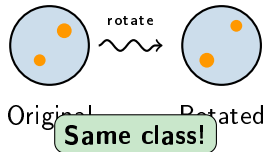
$$f(R \cdot x) = f(x)$$

Same prediction after rotation

Equivariance (for features)

$$f(R \cdot x) = R \cdot f(x)$$

Features rotate coherently



▶ rotation_invariance.gif

The $E(3)$ Group: 3D Euclidean Symmetries

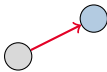
$E(3)$ = Euclidean group in 3 dimensions

Rotations



Any 3D rotation
around any axis

Translations



Shifting the entire
structure in space

Reflections



Mirror
symmetries

Key Insight

We can build architectures that **guarantee** these invariances by construction — not learned through augmentation, but **mathematically proven**.

EGNN: Equivariant Graph Neural Network

Satorras et al., 2021

Key Principle: Use only **invariant quantities** in messages

Message (uses distance = invariant):

$$m_{ij} = \phi_m(h_i, h_j, \|x_i - x_j\|^2)$$

Coordinate update (equivariant):

$$x_i^{new} = x_i + \sum_j (x_i - x_j) \cdot \phi_x(m_{ij})$$

Advantages:

- ✓ No augmentation needed
- ✓ Guaranteed robustness
- ✓ Better with less data

Trade-off:

Slightly lower raw accuracy

Architecture Comparison Summary

Architecture	Aggregation	E(3)-Equiv.	Strength
GCN	Mean	✗	Simple baseline
GAT	Attention	✗	Best accuracy
DeepSets	Global sum	✗	No local structure
EGNN	Distance-based	✓	Guaranteed robustness

Key Trade-off:

- GAT → Best raw performance
- EGNN → Guaranteed geometric invariance

Part II Summary: GNN Theory

- ① **Message passing** enables learning on graphs
- ② **GAT** with attention achieves best accuracy
- ③ **EGNN** provides guaranteed $E(3)$ -equivariance
- ④ **All architectures** outperform spatial statistics on variable morphologies

Next

Real data finally arrives — how to build the processing pipeline?

Part III

From Real Images to Graphs

Processing Pipeline

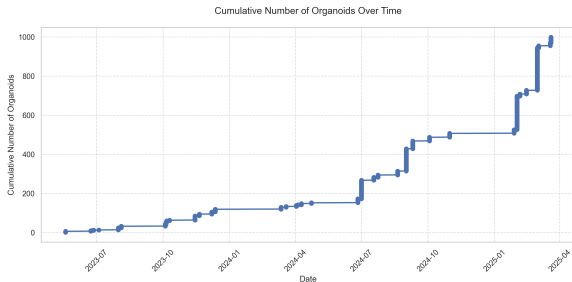
The Real Dataset Arrives

ANR Morpheus Project

- 22 months of data collection
- May 2023 – February 2025
- IPMC Nice + Paris Cité

Dataset Statistics:

- 1,311 imaged samples
- 2,272 extracted organoids
- **500 well-differentiated** selected
- ~250 per class

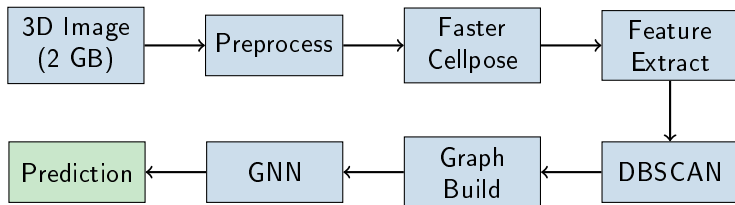


Data accumulated over 22 months

Challenge

How to transform 2 GB 3D images into graphs for GNN processing?

End-to-End Processing Pipeline



Processing time: ~ 20 min/organoid
(dominated by segmentation)

Compression: 2 GB $\rightarrow$ 10 MB
(1000 $\times$ reduction)

Faster Cellpose: Segmentation Bottleneck

Problem: Cellpose is slow

- $30 \text{ sec/slice} \times 200 \text{ slices} \times 1000 \text{ organoids} = \mathbf{2,500 \text{ hours}}$

Our Solution: Faster Cellpose

- 1 **Knowledge Distillation:** Smaller student network (−50% params)
- 2 **Weight Pruning:** Remove 30% low-magnitude weights
- 3 **Mixed Precision:** FP16 inference

Method	F1	Time/Slice	Speed-up
Geometric (ellipses)	0.88	3 sec	10×
Faster Cellpose	0.95	6 sec	5×
Original Cellpose	0.98	30 sec	1×

Graph Construction from Segmented Cells

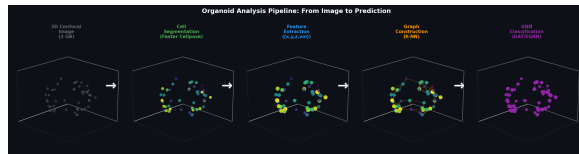
Node Features (4D):

- 3D centroid: (x, y, z)
- Cell volume
- Normalized (centered, scaled)

Edge Construction:

- K-nearest neighbors ($K = 10$)
- Euclidean distance
- Symmetrized + radial cutoff

▶ graph_construction.gif



From segmentation to graph

Part III Summary: Pipeline

- ① **Real dataset:** 500 well-differentiated organoids (22 months)
- ② **Faster Cellpose:** $5\times$ speedup while maintaining $F1 = 0.95$
- ③ **Graph construction:** $1000\times$ compression preserving structure
- ④ **Complete pipeline:** 3D image $\rightarrow$ graph $\rightarrow$ prediction

Next

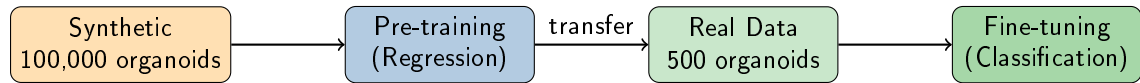
Transfer learning: can synthetic pre-training help with real data?

Part IV

Results & Transfer Learning

From Synthetic to Real

Transfer Learning: Synthetic $\rightarrow$ Real



Phase 1: Pre-training

- 70,000 synthetic organoids
- Task: Regress Matérn parent number
- Learn spatial organization patterns

Phase 2: Fine-tuning

- 500 real organoids
- Reinitialize classification head
- Reduced learning rate (10^{-4})

Performance on Synthetic Data

Task: Regression of Matérn parent number

Test set: 15,000 synthetic organoids

Architecture	MSE	vs GCN
GCN (baseline)	0.198	—
DeepSets	0.145	+27%
EGNN	0.137	+31%
GAT	0.118	+40%

Key findings:

- GAT wins: attention learns which neighbors matter
- EGNN: equivariance divides MSE by **2.8×**

Why GAT wins:

- Attention learns **which neighbors matter**
- Adaptive to clustering patterns
- Multi-head captures different relationships

DeepSets' good performance:

- Global statistics are informative
- But local structure still helps
- GAT > DeepSets confirms this

EGNN trade-off:

- Slightly behind GAT in raw MSE
- But: guaranteed equivariance

Equivariance Benefit

Without equivariance: $\text{MSE} = 0.38$

With EGNN: $\text{MSE} = 0.137$

⇒ **2.8× improvement**

Dataset: 500 well-differentiated organoids

Evaluation: 5-fold cross-validation

84% Accuracy
(GAT + pre-training)

Cauliflower:

- Precision: 93%
- Recall: 74%

Cystic:

- Precision: 78%
- Recall: 95%

Main confusion: Low-deformation cauliflowers → quasi-spherical

Confusion Matrix Analysis

	Pred. Cauliflower	Pred. Cyst.
Cauliflower	28	10
Cystic	2	35

Results:

- 63/75 correct = **84%**
- TP Cauliflower: 28
- TP Cystic: 35

Main Error Pattern:

Cauliflower → Cystic (10 cases)

Interpretation:

- Low-deformation cauliflowers
- Quasi-spherical appearance
- Possible transitional phenotypes

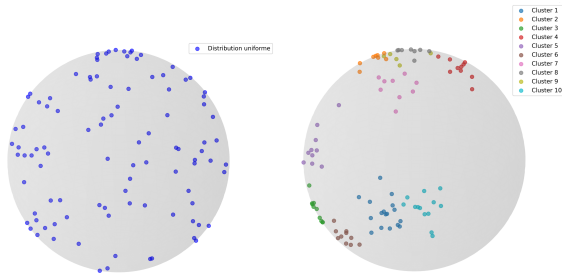
Transfer Learning Impact

Strategy	Accuracy
GAT from scratch	76%
GAT pre-trained	84%
<i>Improvement</i>	+8%

Data Efficiency:

- Pre-trained + 25% data: 78%
- From-scratch + 100%: 76%

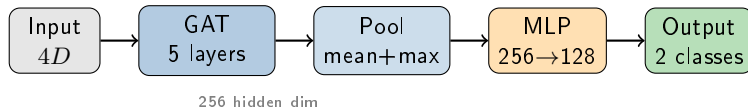
⇒ **4× fewer annotations needed!**



Data efficiency comparison

Convergence: 3× faster

Final Architecture



Training

Optimizer	AdamW
LR (pre-train)	10^{-3}
LR (fine-tune)	10^{-4}

Inference

Speed	200+ org/min
Memory	~8 GB GPU
Hardware	RTX 3080

Learning Curves: Detailed Analysis

Data Efficiency Gains:

Data %	From-scratch	Pre-trained
10%	58%	71% (+13%)
25%	67%	78% (+11%)
50%	72%	82% (+10%)
100%	76%	84% (+8%)

Key Observation:

Largest gains at **low data** regime

Convergence Speed:

- Pre-trained: **20-30 epochs**
- From-scratch: 80-100 epochs
- $\Rightarrow$ **3 $\times$ faster**

Implication:

Synthetic pre-training teaches useful representations of spatial organization

Practical Impact:

125 real organoids (pre-trained) $\approx$
500 real organoids (from-scratch)

Computational Efficiency

Inference Performance:

- 200+ organoids/minute
- GPU batch processing
- Enables high-throughput screening

Memory Footprint:

- ~8 GB GPU memory
- Compatible with standard hardware
- RTX 3080 / A100

Reproducibility:

- 100% deterministic
- No inter-observer variability
- Identical results every run

Scalability:

Organoids	Time
100	~30 sec
1,000	~5 min
10,000	~50 min

vs Manual Analysis

Manual: 30 min/organoid $\times$ 1000 = 500 hours

Automated: 5 min total — **6000 $\times$ faster**

Identified Limitations

- **Segmentation dependency:** Errors propagate through pipeline
- **Validation scope:** Prostate organoids only; generalization needs testing
- **No inter-laboratory validation:** Different microscopes, protocols
- **No interpretability analysis:** Which cells drive predictions?

① GAT achieves best performance:

- 84% accuracy on real data
- MSE 0.118 on synthetic data

② Transfer learning is transformative:

- +8% accuracy improvement
- 4× data efficiency
- 3× faster convergence

③ Practical system:

- 200+ organoids/minute inference
- Fully automated, reproducible

Conclusion

Contributions & Perspectives

Contribution 1: Graph-Based Representation

Geometric Graphs for Organoid Modeling

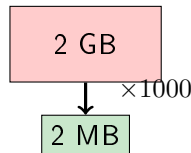
From raw 3D images to compact graph representations

Key Innovation:

- Cell $\rightarrow$ Node abstraction
- Spatial proximity $\rightarrow$ Edges
- 4D features (position + volume)

Impact:

- **1000 $\times$ compression**
- Preserves biological structure
- Enables GNN analysis



Contribution 2: GRETSI Comparative Study

Published at GRETSI 2025

Rigorous comparison: GNNs vs Spatial Statistics

Key Findings:

- GNNs: better **geometric generalization**
- Spatial stats: better **noise robustness**

Recommendation:

GNNs for variable morphologies
Statistics for idealized conditions

Impact:

- Guides method selection
- Justifies deep learning approach
- Foundation for thesis methodology

Citation:

Martin et al., GRETSI 2025

Contribution 3: Synthetic Data Generation

Point Process-Based Generation

Poisson + Matérn processes for synthetic organoids

Method:

- Poisson → Cystic (uniform)
- Matérn → Cauliflower (clustered)

Scale:

- **100,000** synthetic organoids
- Perfect ground truth labels
- Controllable difficulty

Impact:

- Enabled research without real data
- Foundation for transfer learning
- +8% accuracy improvement

Generalizable:

Applicable to other spherical/ellipsoidal biological structures

Contribution 4: Transfer Learning Strategy

Synthetic → Real Transfer

Pre-train on synthetic, fine-tune on real

Results:

+8%

Accuracy improvement

4×

Data efficiency

3×

Faster convergence

Practical Impact

125 organoids (pre-trained) = 500 organoids (from-scratch)

Transforms annotation requirements

Contribution 5: Complete Automated Pipeline

End-to-End System

From raw 3D images to phenotype predictions

Components:

- Preprocessing
- Faster Cellpose (5× speedup)
- Graph construction
- GNN classification

Properties:

- Fully automated
- 200+ organoids/minute
- Open-source
- Reproducible

Code available: github.com/morpheme-inria/organoid-gnn

Methodological Extensions:

- Multi-scale graphs
(cell $\rightarrow$ region $\rightarrow$ organoid)
- Graph Transformers
(global attention mechanism)
- Interpretability analysis
(which cells drive predictions?)

Multi-Modal Integration:

- Spatial transcriptomics + imaging
- Temporal data from time-lapse
- Metabolomic features

Clinical Validation:

- Prospective patient cohorts
- Multi-site testing

Therapeutic Response Prediction:

- Patient-derived organoids
- Test treatments in silico
- Personalized therapy selection

Generative Graph Models:

- In silico organoid design
- Optimize culture protocols
- Computational biology

Multi-Organ Extension:

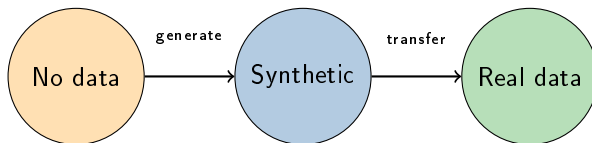
- Brain organoids
- Liver, kidney, lung
- Tumor organoids

Societal Impact:

- **3Rs principles**
Replace, Reduce, Refine
- Accelerate drug discovery
- Reduce animal testing

Geometric Graph Neural Networks

are a powerful tool for 3D organoid analysis



Open-source code available for the community

Thank you for your attention.
I am happy to take your questions.

Backup Slides

Confusion Matrix (Detail)

	Pred. Cauliflower	Pred. Cyst.
Cystic Cauliflower	28	10
	2	35

63/75 correct
= 84% accuracy

Anticipated Questions

① Why GAT over EGNN?

Trade-off: GAT wins on accuracy; EGNN wins on guarantees

② Synthetic data validation?

Empirical gains (+8%) demonstrate practical utility

③ Generalization to other organoids?

Principles transferable; requires fine-tuning Cellpose

④ Why only 500 organoids?

Quality over quantity — ensuring label-morphology consistency

⑤ Clinical deployment?

Would require regulatory validation (IVDR/FDA)

EGNN Formulas (Detail)

Message computation (invariant):

$$m_{ij} = \phi_m(h_i, h_j, \|x_i - x_j\|^2)$$

Coordinate update (equivariant):

$$x_i^{(l+1)} = x_i^{(l)} + \sum_{j \neq i} (x_i - x_j) \cdot \phi_x(m_{ij})$$

Feature update:

$$h_i^{(l+1)} = \phi_h \left(h_i^{(l)}, \sum_{j \in \mathcal{N}(i)} m_{ij} \right)$$

Property

Invariant messages + Equivariant coordinate updates = E(3)-equivariant network

Timing Checkpoints

Section	Target Time
After Introduction (Slide 6)	~5 min
After Part I - Justification (Slide 14)	~15 min
After Part II - GNN Theory (Slide 24)	~25 min
After Part III - Pipeline (Slide 30)	~32 min
After Part IV - Results (Slide 38)	~42 min
End	~45 min

If running long: Abbreviate EGNN details

If running short: Expand perspectives and limitations